

Nishant Bhargava

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nishant7309.github.io/portfolio

EDUCATION

Purdue University

GPA: 3.47/4.0

B.S. in Computer Engineering; Minors in Statistics

Expected May 2028

- **Relevant Coursework:** Multivariate Calculus, Linear Algebra, Statistics, Python for Data Science

PUBLICATIONS

Bhargava, N., et al. "Probe-Rewrite-Evaluate: Quantifying Evaluation Awareness in Large Language Models."
NeurIPS 2025 Workshop on Evaluation for Language Models & Wordplay Workshop @ EMNLP 2025.
[arXiv: 2509.00591](#)

EXPERIENCE

Undergraduate Research Assistant

Sep 2025 – Present

CYNICS Lab @ Purdue

West Lafayette, IN

- Developed a **Transformer**-based linguistic steganography pipeline that perturbs next-token distributions with **<2.0 PPL impact** while embedding encrypted payloads.
- **Fine-tuned** domain **LLMs** for **finance** and **pharma** use-cases, evaluating domain adaptation quality and safety tradeoffs (e.g., perplexity, task accuracy, and leakage risk).
- Designed a password-gated "stego" access mechanism to conditionally reveal and serve hidden fine-tuned model variants only to authorized users (model gating + controlled release).
- Investigating steganographic techniques on **Transformer**-based semantic search, demonstrating persistence of hidden text through vector embedding transformations.

ML Research Lead

Oct 2025 – Feb 2026

The LAB (let all build) Company

Remote

- Developing behavioral analysis models on **AWS SageMaker** by processing chat logs from internal collaboration platforms to quantify team dynamics and predict research bottlenecks.
- Built a custom end-to-end **MLE agent** using a **multi-agent** architecture, scaling from **1 to 2 agent modules** and adding **human-in-the-loop** checkpoints to support ML experiment iteration and technical writeups.
- Supported workflow tracking, documentation, reporting, and coordination across collaborative projects, improving visibility into progress, task ownership, and operational bottlenecks.

AI Safety & Interpretability Researcher

May 2025 – Nov 2025

Algoverse

Remote

- Developed *Probe-Rewrite-Evaluate* methodology to quantify evaluation awareness in **LLMs**, revealing how models alter behavior between test-like and deployment contexts (NeurIPS '25 & EMNLP '25 workshop paper).
- Introduced **Awareness Elasticity** metric to quantify **LLM** sensitivity to evaluation interventions, providing the first systematic framework for measuring benchmark-deployment performance gaps.

Computer Vision Engineer

Jan 2025 – May 2025

RoBoat: Autonomous Maritime Maneuvers

Indianapolis, IN

- Led the AI and perception team for Purdue Indy's autonomous maritime robotics project in the AIMM ICC competition.
- Implemented deep learning-based object detection models (**YOLOv5**) to identify buoys, docks, and obstacles, achieving **93% detection accuracy** in simulation and 88%+ in real-world lake tests.
- Reduced false positives by **34%** through dataset curation, augmentation, and transfer learning on maritime-specific image data.

TECHNICAL SKILLS

Languages: Python, R, C, SQL, Bash, MATLAB

ML/LLM: PyTorch, JAX, scikit-learn, Hugging Face Transformers, transformer-lens, PEFT/LoRA, NumPy, pandas, OpenCV, Weights & Biases

Systems/Tools: Linux, Docker, CUDA, AWS (SageMaker, S3, EC2), Git